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ABSTRACT

Domain-Driven LLM Development: Insights into RAG and Fine-Tuning Practices

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Domain-Driven LLM Development: Insights into RAG and Fine-Tuning Practices

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Abstract

To improve Large Language Model (LLM) performance on domain specific applications, ML developers often leverage Retrieval Augmented Generation (RAG) and LLM Fine-Tuning. RAG extends the capabilities of LLMs to specific domains or an organization's internal knowledge base, without the need to retrain the model. On the other hand, Fine-Tuning approach updates LLM weights with domain-specific data to improve performance on specific tasks. The fine-tuned model is particularly effective to systematically learn new comprehensive knowledge in a specific domain that is not covered by the LLM pre-training. This tutorial walks through the RAG and Fine-Tuning techniques, discusses the insights of their advantages and limitations, and provides best practices of adopting the methodologies for the LLM tasks and use cases. The hands-on labs demonstrate the advanced techniques to optimize the RAG and fine-tuned LLM architecture that handles domain specific LLM tasks. The labs in the tutorial are designed by using a set of open-source python libraries to implement the RAG and fine-tuned LLM architecture.

CCS Concepts

• Compute Methodologies → Artificial Intelligence

Keywords

Retrieval Augmented Generation, Large Language Model, Fine Tuning, Generative Artificial Intelligence

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Summary of the Tutorial

RAG and LLM Fine-Tuning are popular approaches in building LLM applications. In this tutorial we examine the advantages and disadvantages between RAG and fine-tuned LLM in complexity, transparency, cost-effectiveness, time efficiency, and accuracy. We also explore how to combine the two methods in a common LLM pipeline for the domain-specific text generation, question-answer and text summarization tasks.

Lab 1: Advanced Techniques in RAG. In this lab, we explore multi-phase optimization techniques to enhance RAG's accuracy and relevance. The initial RAG struggles with indexing, retrieval, and response generation. Pre-Retrieval Optimization employs advanced chunking and indexing strategies, including semantic chunking and hypothetical question generation. Retrieval Optimization refines user queries and improves retrieval methods, combining keywords and semantic searches with hybrid techniques. Post-retrieval optimization focuses on reranking documents by relevance using heuristic methods and models, along with prompt compression and context condensation. Additionally, Graph RAG uses LLM-generated knowledge graphs for better reasoning across documents, while Agentic RAG supports complex, multi-document analysis. We demonstrate and experiment these optimization techniques for a domain-specific question-answer task in the lab to achieve precise, relevant responses.

Lab 2: LLM Fine-Tuning. This lab focuses on fine tuning LLM using labeled datasets and human/AI feedback to adjust model weights for better performance in specific domains. Fine-Tuning allows smaller, domain-specific models to outperform larger general-purpose models. The Continuous Fine-Tuning process involves Supervised Fine Tuning (SFT) with human-annotated data, followed by human and AI validation, and enhancements via Reinforcement Learning from Human and AI Feedback (RLHF/RLAIF). Integrating fine-tuned LLMs in a RAG pipeline enhances performance by utilizing the most relevant data and learning domain-specific knowledge. In this lab, we first generate synthetic data from domain-specific documents, then

instructively fine-tune a small LLM for the question-answer task previously experimented in Lab 1.

Lab 3: RAG and Fine-Tuned Model Benchmarking. This lab evaluates the performance of RAG, fine-tuned LLM, and a combined architecture on domain-specific question-answer and summarization tasks. We test these approaches using a consistent datasets and performance metrics. Initial metric outcomes guide the decision on whether either approach meets the expected performance. If so, we further estimate the costs associated with each approach, and conduct ROI analysis to determine the most cost-performant solution.

In this hands-on tutorial, we examine the benefits and drawbacks of RAG and LLM Fine-Tuning. With experiments and detailed discussions on the RAG pipeline, fine-tuned LLM, their combined use, and assessment of performance and cost-effectiveness, we demonstrate the best practices to optimize the architecture of RAG and fine-tuned LLM for multiple domain-specific tasks.

Presenters Biography

José Cassio dos Santos Junior A member of the Amazon Machine Learning University team. He is responsible for Curriculum Development for Advanced Modules. As a previous Senior Data Scientist on the AWS LATAM Professional Services Data Science team, he was responsible for experiments and MVP engagements with important clients in the Brazilian market. Cassio has over 20 years of experience working as a software engineer. As a business process management expert, he participated in BPO projects for more than 7 years. Cass also has more than 10 years of teaching experience at colleges and acting as instructor for Linux certification preparation and Microsoft Innovation Center bootcamps. He holds a Master's degree in Computer Engineering, a Bachelor's degree in Physics, and a Bachelor's degree in Business Administration, specialized in IT Quantitative Methods.

Rachel Hu The Co-founder & CEO of CambioML, which builds open-source libraries to prepare high quality data for RAG and LLM finetuning. Previously she was an Applied Scientist at AWS AI, an ML instructor at Amazon Machine Learning University, and a speaker at top conferences including KDD, NVIDIA GTC, AWS re:Invent, MLOps Summit, etc. Rachel co-authored Dive into Deep Learning (D2L.ai), an open-source interactive textbook adopted by over 500 universities around the world. Rachel received her master degree of statistics from University of California, Berkeley, and bachelor of Math from University of Waterloo, Canada.

Richard Song The Co-Founder and CEO of Epsilla Inc, a one-stop RAGaaS platform for building production ready LLM applications. With a background in big data, vector graph databases, and high performance computing, Richard helps customers build production-ready RAG systems connected with large scale proprietary data. Richard holds a Master's degree in Computer Science from Cornell University.

Yunfei Bai A Senior Solutions Architect at Amazon Web Services. With over 15 years' experience on AI/ML, Data Science and Analytics, Yunfei helps AWS customers adopt AI/ML and Generative AI services to deliver business results. Prior to AWS, he worked in various roles including product manager and solution consultant in multiple industries, designed and delivered AI/ML and data analytics solutions that overcome complex technical challenges and drive strategic objectives. Yunfei has a PhD in Electronic and Electrical Engineering. He has published research papers and blog posts, and serves as a journal reviewer.

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